



Photographs provided by Fukuoka City

11th IEEE Electron Devices Technology and Manufacturing (EDTM) Conference

Fukuoka, Japan, March 8th- 11th, 2027

<https://attend.ieee.org/edtm-2027/>



*Conference Theme: “Empowering the AI Era:
The Role of Innovation in Semiconductor Devices and Manufacturing”
Three-page camera-ready paper submission deadline: Mid-October, 2026
Notification for Acceptance: Late December, 2026
Conference format: Please visit website for details.*

IEEE Electron Devices Technology and Manufacturing (EDTM)

11th EDTM (Electron Devices Technology and Manufacturing) conference is a full four-day meeting to be held at Fukuoka, Japan from March 8th to 11th, 2027, fully sponsored by the IEEE Electron Devices Society (EDS). As challenges in semiconductor technology scaling continue to grow, the industry's collaborative efforts must increase to overcome them. EDTM is intended to serve as a forum for the electron devices community to collaborate on topics ranging from devices, materials, and tools, to create new and innovative technologies. EDTM will provide the following new formats.

Technical Areas

EDTM 2027 solicits papers in all areas of electron devices, including materials, processes, devices, memories, photonics, power/RF devices, modeling, reliability, packaging, sensors, flexible electronics, nanotechnologies and neuromorphic/quantum technologies. Authors should select a technical category based on the detailed descriptions in this flyer during online submission of papers.

Oral and Poster Sessions

EDTM 2027 will include parallel technical sessions of oral and poster presentations. Best Paper Award, Best Student Paper Awards and Best Poster Awards will be selected.

Publications

EDTM 2027 papers will be subjected to IEEE standard review processes and publishing guidelines. The accepted and presented papers will be published in the **EDTM 2027** Proceedings included in *IEEE Xplore*. The authors of a selected number of high-impact papers will be invited to submit extended versions for publication in the *IEEE Journal of Electron Devices Society* (J-EDS), subjected to J-EDS review and publication policy.

Short Courses and Tutorials

EDTM 2027 will start with a set of short courses and tutorials on March 8th, 2027. Tutorials teach selected topics from the basics to the state-of-the-arts, allowing the attendees to catch up a topic quickly. Short Courses discuss the latest research and challenges on hot and advanced topics encompassing the EDTM 2027 Theme.

Exhibition

EDTM 2027 offers vendors to showcase their newest products and technologies, allowing attendees to learn about new tools and techniques that can help them to succeed at work.

General Chair:

Masumi Saitoh (Kioxia)

TPC Chair:

Kensuke Ota

(Sony Semiconductor Solutions)

Steering Committee:

Shuji Ikeda (Chair, tei Solutions)

Masumi Saitoh (Kioxia)

Arokia Nathan (Cambridge Univ.)

Bin Zhao (Intel)

Ravi M. Tody (Synopsys)

John Dallesasse (UIUC)

Murty Polararapu

(Space Electronics Solutions)

Meikei leong (Simbury Limited)

Subramaniam Iyer (UCLA)

P. Susthitha Menon (UKM)

Chen Yang (Analog Devices)

Merlyne de Souza (Univ. of Sheffield)

Bernard Lim (Appscard)

Yang Chai (Poly Univ.)

Yogesh Chauhan (IIT Kanpur)

Changrock Song (SK Hynix)

Kazunari Ishimaru (Rapidus)

Hitoshi Wakabayashi (Science Tokyo)

Executive Committee:

Shuji Ikeda (tei Solutions)

Hitoshi Wakabayashi (Science Tokyo)

Kazunari Ishimaru (Rapidus)

Masumi Saitoh (Kioxia)

Kensuke Ota

(Sony Semiconductor Solutions)

Yuichiro Ando (Osaka Metropolitan Univ.)

Kimihiro Kato (AIST)

Kazuaki Iwasawa (TEL)

Shoichi Kabuyanagi (Kioxia)

Tadashi Yamaguchi (Renesas)

Takuya Maeda (Univ. Tokyo)

Secretariat:

PCO, Inc.

Email: edtm2027@pcojapan.jp

Original papers solicited in the following, but not limited to, technical areas:

Materials: All device-related materials, including semiconductors, magnetics, ferroelectrics, insulators, metals, liquid crystals, photoresists, organic films, precursors, EUV photoresists, etching gases, filamentary-switching materials, and phase-changing materials. Materials engineering for reducing costs, and improving reliability/yield/manufacturability. Smart materials enabling intelligent devices are highly welcome.

Processes, Tools, Yield, and Manufacturing: Semiconductor processes and equipment, including process modules (deposition, dry/wet etch, cleaning, planarization, isolation, dielectrics, metals, silicides, lithography), process integration, process control, equipment impact on devices, reliability and yield, self-assembly techniques, process sensing, process enhancement by AI/ML, etc.

Advanced Logic Devices: All semiconductor devices including Si/Ge CMOS, gate all around, nanosheets, interconnects, compound semiconductors, oxide semiconductors, low-dimensional nanomaterials (van der Waals heterostructures, nanotubes, quantum dots), ferroelectrics, spintronics, 3D devices, RF/THz devices, etc. Emerging device concepts for future computing are encouraged, such as tunnel FETs, negative capacitance FETs, topological insulators, phase transitions, qubit devices, etc.

Memory Technologies: All memories, including embedded and standalone memories, volatile and nonvolatile memories, in-memory and neuromorphic computing. Topics on charge-based memories, RRAM, MRAM, PCM, FeRAM, cross-point and selectors, bio-inspired memories, scaling, processing, characterization, reliability, modeling, 3D integration, read/write/erase, novel hierarchies and architectures for memory-centric computing/AI.

Photonics, Imaging and Display: Topics on photonics, photonics for energy, optoelectronics, microwave photonics, nano-photonics, optical sensors including emerging materials, perovskites, optical communications/networking, optical switches, bio-photonics, lasers, optical systems, bio-imaging, imagers, display technologies including QLEDs, OLEDs, VR, AR, and others, and other emerging technologies in photonics, imaging and display.

Power, RF, and Energy Devices: Device technologies related to high-voltage devices, power/RF devices, energy harvesting devices, photovoltaics, energy storage devices, discrete/integrated power devices, power modules/systems. Power device structures such as diodes, BJTs, FETs, and IGBTs. Power device materials such as wide bandgap and ultra-wide bandgap materials (SiC, GaN, AlN, GaO, etc.). Power device fabrication processes, modeling and simulation.

Modeling and Simulation: Advances in modeling/simulation of electron devices, packages, and processes. Numerical, analytical, and statistical modeling/simulation of electronic, optical, or hybrid devices, interconnects, technology CAD, benchmarking, monolithic integration, heterogeneous integration, design-technology co-optimization (DTCO), system-technology co-optimization (STCO), parasitic elements, fabrication processes, physical phenomena, mechanical systems, electro-thermal effects, test structures, and methodologies.

Reliability and testing: Advances in reliability and testing of materials, processes, devices, modules, and systems including interconnects, ESD, latch-up, soft errors, radiation, noise and mismatch behavior, hot-carrier effects, self-heating, biasing and thermal instabilities, dielectric wear-out and breakdown, process charging damage, electromigration, reliability test structures and methodologies, defect monitoring and control, electromagnetic robustness, and design-for-reliability. Also, topics on reliability of emerging memories, more-than-Moore applications, biomedical devices, automotive, and aerospace applications.

Packaging and Heterogeneous Integration: Advances in packaging and heterogeneous integration technologies, including xD integrations. Topics on advanced packaging and manufacturing technologies, such as wafer level packaging, chiplets, ultra-fine-pitch interconnection, sub-micron package-level wiring, optical/wireless interconnects, power/sensor device packaging, controlling thermal-expansion coefficient, thermal management.

Sensors, MEMS, Bio-electronics: Advances in sensors and actuators including MEMS, Bio-MEMS/NEMS, transducers, resonators, micro/nano-fluidic devices, bio-sensors, implantable bio-medical devices, biomolecular-based memories, bio-transistors, semiconductor synthetic biological devices/systems, semiconductor synaptic/neural devices, brain-inspired computing, brain-interface devices, and heterogeneous integration with CMOS.

Flexible and Wearable Electronics: Topics on flexible and wearable electronics including flexible and wearable transistors and related integrated circuits, sensors, RFID, light emitting diodes, lighting and display, energy harvesting and storage devices, novel and manufacturing processes for flexible and wearable electronics, and materials for flexible and wearable electronics.

Nanotechnologies: Advances in nanotechnologies including nanomaterials, nanoelectronics, low-dimensional systems including 2D materials and devices, nanophotonics, nanofabrication, nanoenergy, nanobiomedicine, nanosensors, and related nano characterization/modeling techniques.

Disruptive Technologies: Devices and architectures related to metaverse, neuromorphic computing, quantum computing, in-memory, neuromorphic, and bio-inspired computing, AI/ML for next-generation computing, quantum machine learning, and logic-in-memory.

Three-page camera-ready paper submission deadline: Mid-October, 2026
Notification for Acceptance: Late December, 2026